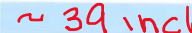


USING METRIC UNITS

The most commonly used measurement system is the **metric system**. The standard metric unit of length is the **meter**. Some of the most common metric units of length are shown here with similar customary units.

Metric Units of Length	
1 millimeter (mm)  ✓	1 meter (m)  ~ 39 inches
$\frac{1}{16}$ inch (in)  actual size	1 yard (yd)  to scale ✓
1 centimeter (cm)  ✓	1 kilometer (km)  ✓
1 inch (in)  actual size	1 mile (mi)  to scale

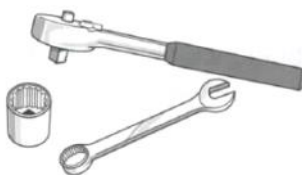
10 millimeters = 1 centimeter
 1,000 millimeters = 1 meter
 100 centimeters = 1 meter
 1,000 meters = 1 kilometer

american

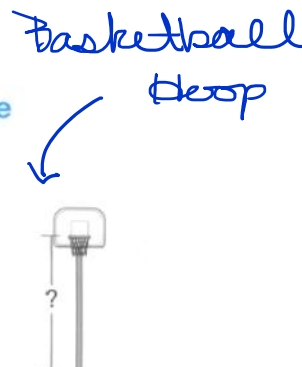
Choose the units you would use from the **customary** and metric systems to measure each of the following items.



1. Kilometers
miles



2. Centimeter
(possibly millimeters)
inches



3. Meters
feet or yards

Metric Length, cont'd...

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10 millimeters = 1 centimeter

~~1,000 millimeters = 1 meter~~

100 centimeters = 1 meter

1,000 meters = 1 kilometer

Do you see the pattern?

1 kilometer = 1000 meters → multiply by 1000

1 meter = 100 centimeters → multiply by 100

1 centimeter = 10 millimeters → multiply by 10

Do you see how looking for a pattern helps? Now we know that as we go from large to small, our multiplier decreases by a factor of 10. No more need to memorize!!

Look at the chart above. Notice that the metric system is based on tens. For example, 10 millimeters equal 1 centimeter and 100 centimeters equal 1 meter.

When you are converting metric measurements, remember that the metric system is based on tens.

To change a larger metric unit to a smaller unit, multiply by the number of smaller units in the larger unit.

Example 1.5 km = _____ m

Think: $1.5 \times 1,000 = 1,500 = 1,500 \text{ m}$

→→→
Move decimal point 3 places.

$$1 \text{ km} = 1,000 \text{ m}$$

$$1.5 (1 \text{ km}) = 1.5 (1,000 \text{ m})$$

$$1.5 \text{ km} = 1,500 \text{ m}$$

writing out the formulas helps to learn the formulas

Change the following measurements to the units indicated.

4. 19.4 cm = 194 mm

$$(19.4) 1 \text{ cm} = 10 \text{ mm}$$

$$19.4 \text{ cm} = 194 \text{ mm}$$

6. 20 km = 20,000 m

$$(20) 1 \text{ km} = 1,000 \text{ m}$$

$$20 \text{ km} = 20,000 \text{ m}$$

8. .5 km = 500 m

$$(\frac{1}{2}) 1 \text{ km} = 1,000 \text{ m}$$

$$.5 \text{ km} = 500 \text{ m}$$

5. 8.7 m = 870 cm

$$(8.7) 1 \text{ m} = 100 \text{ cm}$$

$$8.7 \text{ m} = 870 \text{ cm}$$

7. 2.9 cm = 29 mm

$$(2.9) 1 \text{ cm} = 10 \text{ mm}$$

$$2.9 \text{ cm} = 29 \text{ mm}$$

9. .64 m = 64 cm

$$(.64) 1 \text{ m} = 100 \text{ cm}$$

$$.64 \text{ m} = 64 \text{ cm}$$

Metric Capacity & Weight

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Some common metric units of weight and capacity are shown below. The standard metric unit of weight is the **gram**. The standard metric unit of capacity is the **liter**. Remember, capacity measures liquid or granular amounts.

Metric Units of Weight	
1 milligram (mg)	✓ about the weight of a grain of salt
1 gram (g)	✓ about the weight of a pencil eraser $28\text{ g} \approx 1\text{ oz}$
1 kilogram (kg)	✓ a little over 2 pounds $50\text{ kg} \approx 100\text{ lbs}$

Metric Units of Capacity	
1 milliliter (ml)	about the size of a sugar cube
1 liter	a little more than 1 quart

1,000 milligrams = 1 gram
1,000 grams = 1 kilogram
1,000 milliliters = 1 liter

$$1\text{ km} = 1000\text{ g} \quad \times 1000$$

$$1\text{ g} = 1000\text{ mg} \quad \times 1000$$

∴ weight - factors of 1000
capacity - factor of 1000
american

Choose the units you would use from the customary and metric systems to measure each of the following items.



10.

kilograms
pounds



11.

liter or
grams
ounces



12.

liter
quarts

Metric Capacity & Weight, cont'd...

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To convert smaller metric units to larger units, reverse the process.

To change a smaller metric unit to a larger unit, *divide by the number of smaller units in the larger unit.*

Example 2,500 g = 2.5 kg *Factor of 1000; therefore 2.5 kg*

Think: $2,500 \div 1,000 = 2,500 = 2.5 \text{ kg}$

Move decimal point 3 places.

Not the best method for remembering !!

Change the following measurements to the units indicated.

13. 500 mg = .5 g

15. 22 ml = .022 liter

17. 800 g = 0.8 kg

$$\begin{aligned} \left(\frac{1}{2}\right) 1000 \text{ mg} &= 1 \text{ g} \left(\frac{1}{2}\right) & \left(\frac{22}{1000}\right) 1000 \text{ mL} &= 1 \text{ L} \left(\frac{22}{1000}\right) & \left(\frac{800}{1000}\right) 1000 \text{ g} &= 1 \text{ kg} \left(\frac{8}{10}\right) \\ 500 \text{ mg} &= .5 \text{ g} & 22 \text{ mL} &= 0.022 & 800 \text{ g} &= 0.8 \text{ kg} \end{aligned}$$

14. 1,200 ml = 1.2 liters

16. 4,500 g = 4.5 kg

18. 25 mg = 0.025 g

$$\begin{aligned} \left(\frac{1.2}{1}\right) 1000 \text{ mL} &= 1 \text{ L} \left(\frac{1.2}{1}\right) & \left(\frac{4.5}{1}\right) 1000 \text{ g} &= 1 \text{ kg} \left(\frac{4.5}{1}\right) & \left(\frac{25}{1000}\right) 1000 \text{ mg} &= 1 \text{ g} \left(\frac{25}{1000}\right) \\ 1200 \text{ mL} &= 1.2 \text{ L} & 4500 \text{ g} &= 4.5 \text{ kg} & 25 \text{ mg} &= 0.025 \end{aligned}$$

too complicated
to Remember!!

MAKING CONVERSIONS

Your work with ratio and proportion can help you with more complex measurement conversions. You can use unit multipliers to perform several conversions at once. A **unit multiplier** is a fraction, written with two different measurement units, that has a value of one.

Examples of Unit Multipliers

$$\frac{12 \text{ inches}}{1 \text{ foot}} \quad \frac{1 \text{ pound}}{16 \text{ ounces}} \quad \frac{100 \text{ centimeters}}{1 \text{ meter}}$$

Can you see that each unit multiplier equals 1? In each case, the value in the numerator is equal to the value in the denominator.

this I can remember:

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ m} = 100 \text{ cm}$$

$$1 \text{ cm} = 10 \text{ mm}$$

Notice
× 1000
× 100
× 10

USING UNIT MULTIPLIERS

Example How many centimeters are in 3 kilometers?

Step 1

Write the measurement you are converting as a fraction over 1.

$$\frac{3 \text{ kilometers}}{1}$$

Step 2

Make a chain of unit multipliers using the facts that you know.

You know that 1 km = 1,000 m and 100 cm = 1 m.

$$\frac{3 \text{ kilometers}}{1} \times \frac{1,000 \text{ meters}}{1 \text{ kilometer}} \times \frac{100 \text{ centimeters}}{1 \text{ meter}}$$

Step 3

Multiply the expression. You can cancel like measurement units just as you would cancel factors when multiplying fractions.

$$\frac{3 \text{ km}}{1} \times \frac{1,000 \text{ m}}{1 \text{ km}} \times \frac{100 \text{ cm}}{1 \text{ m}} = \frac{3 \times 1,000 \times 100 \text{ cm}}{1} = 300,000 \text{ cm}$$

There are 300,000 centimeters in 3 kilometers.

This I cannot:

Now ...

$$1 \text{ km} = 1000 \text{ m} \\ = 1000 (100 \text{ cm}) \\ = 100000$$

$$3 \text{ km} = 300000$$

Use unit multipliers to make the conversions.

1. How many fluid ounces are in 1 quart?

$$\frac{1 \text{ qt}}{1} \times \frac{2 \text{ pt}}{1 \text{ qt}} \times \frac{2 \text{ c}}{1 \text{ pt}} \times \frac{8 \text{ fl oz}}{1 \text{ c}} = 32 \text{ fl oz}$$

$$1 \text{ quart} = 32 \text{ fl oz}$$

2. How many milligrams are in 1 kilogram?

$$\frac{1 \text{ kg}}{1} \times \frac{1,000 \text{ g}}{1 \text{ kg}} \times \frac{1,000 \text{ mg}}{1 \text{ g}} = 1,000,000 \text{ mg}$$

$$1 \text{ kg} = 1000 \text{ g} \\ 1 \text{ g} = 1000 \text{ mg}$$

$$\therefore 1 \text{ kg} = 1000 (1000 \text{ mg}) \\ = 1,000,000 \text{ mg}$$

Conversions, cont'd...

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On the GED math test, you will be expected to know the basic conversions that you have learned in this unit. If any other measurement fact is needed to solve a problem, the information will be provided with the test item. You can create a unit multiplier from any fact that you are given.

USING NEW MEASUREMENT FACTS

Example Justin needs to know how many inches are equal to 1 meter. He is given the following information: $1 \text{ in} = 2.54 \text{ cm}$.

Step 1

Justin needs to write a chain of unit multipliers that connects meters and inches. In addition to $1 \text{ in} = 2.54 \text{ cm}$, he needs another fact that connects centimeters and meters. He uses the fact that $100 \text{ cm} = 1 \text{ m}$.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \text{ cm}}{1 \text{ m}}$$

There are about **39.37 inches** in 1 meter.

Step 2

Multiply. Cancel like measurement units.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \text{ cm}}{1 \text{ m}} = \frac{100 \text{ in}}{2.54 \text{ m}}$$

Step 3

Divide to create a unit rate.

$$100 \div 2.54 \approx 39.37 \text{ inches per meter}$$

$$\frac{1 \text{ in} = 2.54 \text{ cm}}{2.54 \quad 2.54}$$

$$(100) \quad 0.3937 \text{ in} = 1 \text{ cm} \quad (100)$$

$$39.37 \text{ in} = 100 \text{ cm}$$

$$100 \text{ cm} = 1 \text{ m}$$

$$\therefore 39.37 \text{ in} = 1 \text{ m}$$

Solve the following problems.

3. A chemist knows that there are **16 drams in 1 ounce**. How many drams would there be in **2 pounds**?

$$\begin{aligned} (32) \quad 16 \text{ drams} &= 1 \text{ ounce} \\ &\rightarrow 2 \text{ pounds} = 32 \text{ oz} \\ &\rightarrow 32 \text{ oz} = 512 \text{ drams} \end{aligned}$$

4. How many seconds are in 4 hours?

$$\begin{aligned} (60) \quad 60 \text{ sec} &= 1 \text{ min} \quad (60) \\ &\rightarrow 60 \text{ min} = 1 \text{ hr} \\ &\rightarrow 3600 \text{ sec} = 60 \text{ min} \\ (4) \quad 3600 \text{ sec} &= 1 \text{ hr} \quad (4) \\ 14400 \text{ sec} &= 4 \text{ hrs} \end{aligned}$$

6. One liter is equal to about 1.06 quarts. There are 4 quarts in 1 gallon. Rounded to the hundredths place, about how many liters are in 1 gallon?

$$\begin{aligned} (\div 1.06) \quad 1 \text{ L} &= 1.06 \text{ qts} \quad (\div 1.06) \\ &\rightarrow 4 \text{ qt} = 1 \text{ gal} \\ &\rightarrow (4) \quad 3.7736 \text{ L} = 1 \text{ qt} \quad (4) \\ &\rightarrow 3.7736 \text{ L} = 4 \text{ qts} \\ &\rightarrow 3.7736 \text{ L} = 1 \text{ gal} \\ &\rightarrow 3.77 \text{ L} \end{aligned}$$

7. One mile equals approximately 1.6 kilometers. In miles, about how long is a 10-km race?

$$\begin{aligned} \div 1.6 \quad 1 \text{ mi} &= 1.6 \text{ km} \quad \div 1.6 \\ (10) \quad 0.625 \text{ mi} &= 1 \text{ km} \quad (10) \\ 6.25 \text{ mi} &= 10 \text{ km} \end{aligned}$$

5. In a history book, you read about a statue that weighs 5 stones. You find out that 1 stone is equal to 224 ounces. How many pounds does the statue weigh?

(Use the fact: 1 pound = 16 ounces.)

- A. 14
B. 44.8
C. 70
D. 1,120

$$\begin{aligned} (5) \quad 1 \text{ stone} &= 224 \text{ oz} \\ 5 \text{ stone} &= 1120 \text{ oz} \\ 16 \text{ oz} &= 1 \text{ lb} \\ 1120 / 16 &= 70 \text{ lbs} \end{aligned}$$

Note, the English know
1 stone = 14 lbs.
 $\therefore 5 \text{ stone} = 70 \text{ lbs.}$

8. In the metric system, there are units that are less often used in the United States. For example, a hectogram is equal to 0.1 kilogram. How many grams are in a hectogram?

- A. 10
B. 100
C. 1,000
D. 10,000

$$\begin{aligned} 1 \text{ hg} &= 0.1 \text{ kg} \\ \therefore 10 \quad 1 \text{ kg} &= 1000 \text{ g} \div 10 \\ 0.1 \text{ kg} &= 100 \text{ g} \\ 100 \text{ g} &= 1 \text{ hg} \end{aligned}$$